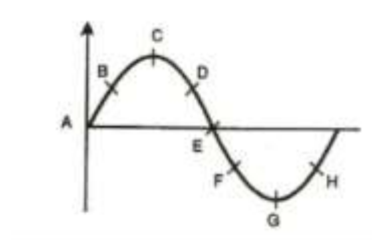
<u>Physics Assignment -2</u>

1. A wave is represented by $y(x,t) = [8 \text{ cm}] \sin[(10 \text{ rad/cm})\, x - (10 \text{ rad/s})\, t]$
   Determine the amplitude, wavelength, angular frequency, wave number, and wave velocity.
2. A transverse wave travels along a string from left to right. The figure below represents the shape of the string at a given instant. At this instant
   (a) Which points have an upward velocity?
   (b) Which points have a downward velocity?
   (c) Which points have zero velocity?
   (d) Which points have the maximum magnitude of velocity?



3. A 1-D sinusoidal wave is propagating along the positive x-direction. The displacement at two points x1 = 0 and x2 = 2.0 cm is given by the following expressions:
   $$Y(x1, t) = (0.02 \text{ cm}) \sin[(3\pi \text{ s-1})\, t]$$
   $$Y(x2, t) = (0.02 \text{ cm}) \sin[(3\pi \text{ s-1})t + \pi/2]$$
   Determine the amplitude, frequency, wavelength, direction of propagation, and speed of the wave.
4. Show that the following functions satisfy the wave equation
   a) $y(x, t) = (x - vt)^2$
   b) $y(x, t) = a \sin(kx - \omega t)$
   c) $y(x, t) = a \sin(kx + \omega t)$
   d) $y(x, t) = a\, e^{i(kx - \omega t)}$
5. Transverse waves are generated in two steel wires A & B of diameters 0.001 m and 0.0005 m, respectively, by attaching their free end to a vibrating source of frequency 500 Hz. Find the ratio of the wavelength if they are stretched with same tension.
6. Two coherent waves of the same frequency and constant phase difference having intensities I and 4I, respectively, interfere. Calculate the resultant intensity when the phase difference between these two waves is $\pi/2$ and π.
7. Two waves of the same frequency and constant phase difference have intensities in the ratio 81:1. They produce interference fringes. Deduce the ratio of the maximum to minimum intensity.
8. In Newton's Rings experiment, the diameter of the 15th ring was found to be 0.59 cm, and that of the 5th ring was 0.336 cm. If the radius of the plano-convex lens is 100 cm, calculate the wavelength of light used. What happens to ring diameter if air film is replaced with liquid of refractive index 1.5?
9. A thin glass sheet of refractive index (μ) 1.5 is introduced normally in the path of one of the two interfering waves. The central bright fringe is observed to shift to the position originally occupied by the eight bright fringe using a monochromatic source which emit light of wavelength (λ) 5890 Å. Determine the thickness of glass sheet.

10. In newton's rings experiment if the radius of curvature of plano convex lens is 200 cm and wavelength of the light used is 5890 Å, Calculate the diameter of 10^{th} bright ring.
11. In Newton's ring experiment two light sources of wavelength 6000Å and 4500 Å are used to form rings. It is observed that n^{th} dark ring due to 6000 Å light coincide with $(n+1)^{th}$ dark ring due to 4500 Å. If the radius of curvature of the plano convex lens is 100 cm, calculate the diameter of nth dark ring due to 6000Å and 4500 Å.
12. Show that the width of the principal maximum in the diffraction pattern of a single slit is twice the width of secondary maxima.
13. In the experimental setup for obtaining the Fraunhofer diffraction pattern of a vertical slit of width 0.3 mm, the focal length of the lens kept between the slit and the screen is 30 cm. Calculate (a) the diffraction angles and positions of the first, second, and third minima, and (b) the positions of the first, second, and third secondary maxima on either side of the central spot. The slit is illuminated with yellow sodium light, which is a doublet. You take $\lambda = 6000$ Å.
14. In the experimental setup given above in Q.13, we change slit widths to 0.2 mm, 0.1 mm, and 0.06 mm. Calculate the positions of the first and second minima.
15. In the previous experimental setup as mentioned in Q.13, consider that slit width b has values $b = 10\lambda$, 5λ, and λ. Calculate the spread and plot the central maximum for each value of slit width.

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